Final Report

What Factors Affect Buldak Sauce Stain Removal Efficiency on White Cotton Fabric?

Design of Experiments — Final Report | Team 4

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1. Research Question and Objective

1.1 Research Question

This study was motivated by a practical problem common in everyday life: the difficulty of removing food stains from clothing. South Koreans frequently encounter red-pigmented dishes — kimchi, Malatang, and Buldak sauce — that easily discolor fabrics and are notoriously resistant to laundering. Identifying effective removal conditions could prevent the need to discard garments or resort to costly professional cleaning services.

Buldak sauce was selected as the target contaminant, inspired by its surging domestic and international popularity. Because it is a complex emulsion containing chicken extract, chili oil, soybean oil, and various spices, it presents a challenging and practically relevant stain whose removal may depend on the chemical properties of the detergent, water temperature, and soaking duration.

Research question: Which combination of detergent type, water temperature, and soaking time most effectively removes Buldak sauce stains from white cotton fabric, as quantified by CIEDE2000 color difference (ΔE^*_{00})?

Feasibility of answering: Because each of the eight treatment combinations is directly measured, the experiment's outcome — the set of cell means — directly identifies which specific combination minimizes ΔE^*_{00}, answering the research question without relying on extrapolation. Moreover, because the design estimates interactions rather than only main effects, it can reveal conditional optima (e.g., the best soaking time depends on temperature) that a one-factor-at-a-time approach would miss — precisely the kind of result this experiment produced.

1.2 Experimental Objective

The experiment evaluates the main effects and interaction effects of three factors on stain removal efficiency:

- **Factor A:** Detergent type (dishwashing detergent vs. laundry detergent)
- **Factor B:** Water temperature (25°C ambient vs. 60°C heated)
- **Factor C:** Soaking time (5 min vs. 30 min)

Response variable: ΔE^*_{00} (CIEDE2000 color difference between the fabric before contamination and after laundering). A smaller ΔE^*_{00} indicates that the post-wash fabric color is closer to its original white state, corresponding to more effective stain removal.

2. Experimental Design

2.1 Treatment Definitions

Factor A — Detergent Type

- **Dishwashing detergent (D):** Saenghwal Gongjakso brand dishwashing liquid. 5 g weighed on a digital scale and fully dissolved in 342.4 g of room-temperature tap water (25°C), yielding a 1.44% w/w solution.
- **Laundry detergent (L):** Saenghwal Gongjakso brand liquid laundry detergent. 5 g dissolved in 342.4 g of room-temperature tap water (25°C) under the same protocol.

Rationale: The main components of Buldak sauce — chili oil and soybean oil — are lipid-based stains. Dishwashing detergents are specifically formulated to emulsify such lipid-based stains, whereas laundry detergents are specialized for removing stains fixed to fabric. We expect the comparison to yield interesting results given their different surfactant chemistries.

Factor B — Water Temperature

- **Ambient (25°C):** Standard tap water. Temperature verified with a mercury thermometer immediately before immersion. Trials where temperature falls outside 23–27°C are discarded and repeated.
- **Heated (60°C):** Water boiled in an electric kettle and cooled to 60°C. Temperature measured immediately before immersion. Trials outside 58–62°C are discarded and repeated.

Rationale: The experimenter's washing machine offers settings of 30°C, 40°C, 60°C, and 95°C. The 95°C setting was excluded (risk of fabric damage); 30°C was excluded (negligible difference from 25°C tap water); 40°C was excluded (cools to near-ambient too rapidly during a 30-minute soak). A pilot run confirmed that 60°C is stable within ±2°C throughout a 30-minute soak using a thermally insulated container.

Factor C — Soaking Time

- **Short (5 min):** Representing an immediate wash under time constraints. Soaking begins precisely 30 minutes after sauce application and lasts exactly 5 minutes, timed by stopwatch.
- **Long (30 min):** Representing a deliberate pre-soak. Soaking begins at the same elapsed time and lasts 30 minutes.

Rationale: Soaking expands fabric fibers and softens fixed contaminant, increasing laundering efficiency. The two levels contrast an immediate time-constrained wash (5 min) with a thorough pre-soak (30 min).

2.2 Response Variable

Response variable: ΔE^*_{00} (CIEDE2000 color difference). Unit of measurement: CIEDE2000 units.

CIEDE2000 is the current international standard for perceptually uniform color difference, incorporating corrections for lightness, chroma, and hue. Unlike simple mean RGB brightness or ΔL^*, it captures the full three-dimensional color change caused by laundering, including the reddish-orange hue of Buldak sauce. A smaller ΔE^*_{00} indicates a smaller difference between the pre-contamination and post-wash states, corresponding to more effective stain removal.

Perceptual reference values: $\Delta E^*_{00} < 1.0$ = imperceptible; 1.0–2.0 = barely perceptible; 2.0–3.5 = clearly perceptible; > 3.5 = large difference.

In this study, a difference of $\Delta E^*_{00} \geq 2.0$ between two treatment conditions is treated as practically meaningful, as this corresponds to the threshold under the CIEDE2000 standard at which a color difference becomes clearly perceptible to the human eye. This criterion is used both as the basis for sizing the study in the power analysis (Section 5) and as the standard for interpreting the practical significance of the final results in the conclusions (Section 9).

Photography protocol:

- **Camera:** Smartphone in Pro mode with white balance and exposure locked throughout all trials.
- **Location:** Bathroom with door fully closed to block ambient light; single standardized overhead LED source.
- **Setup:** Paper foil on toilet lid, cutting board on top; experimenter's elbow rested on towel rack to fix camera height.
- **Distance:** Camera to cutting board fixed at 100 cm for all trials.
- **Timing:** All photographs taken at the same time of day to ensure consistent ambient conditions. Photos taken (i) immediately before stain application (pre-wash baseline) and (ii) after drying the following day (post-wash), under identical conditions.

Calibration procedure:

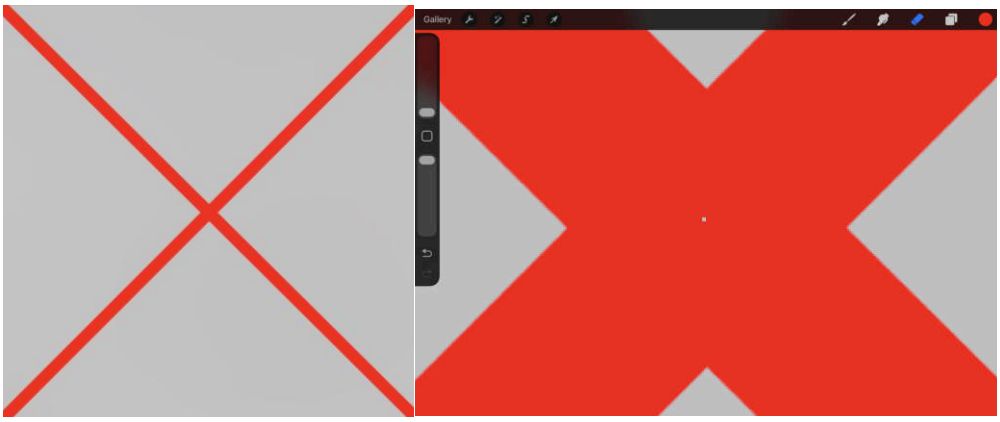
To correct for microscopic between-session lighting variation, a fixed calibration anchor point was established in every image: the center of the non-slip pad on the bottom-left corner of the cutting board. Both pre- and post-wash photos were digitally adjusted until the anchor-point color matched the reference value.

Calibration Point	RGB	CIE L*a*b*
Pre-contamination reference	(153, 152, 148)	L*=62.82 (a*=−0.39, b*=2.23)
Post-contamination reference	(150, 149, 147)	L*=61.73 (a*=−0.02, b*=1.18)

ΔE^*_{00} computation (R):

1. Apply Gaussian blur to the central fabric region of the calibrated image to smooth pixel noise.


2. Draw red guide lines connecting the four fabric corners; extract mean RGB from the central region of interest (ROI).



3. Convert RGB to CIE L*a*b* using `farver::convert_colour(from="rgb", to="lab")`.
4. Compute CIEDE2000 per observation using `farver::compare_colour(from_space="lab", method="cie2000")`.

2.3 Experimental Units and Design Structure

Experimental units: white cotton fabric pieces cut uniformly to 5 cm × 5 cm. Day 1 pieces were cut from the same white cotton T-shirt; Day 2 pieces were cut from a new fabric after the original fabric was exhausted. Pieces were shuffled to destroy positional information and numbered sequentially (1-1, 1-2, 2-1, …) with permanent marker at a corner outside the ROI.

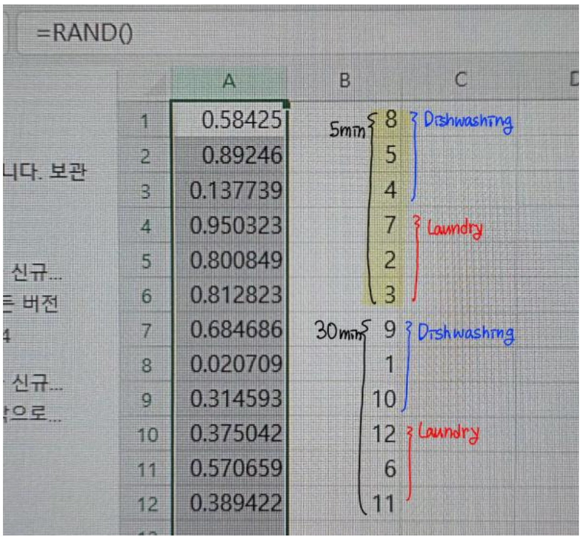
Design: 2^3 full factorial (3 two-level factors, all 8 treatment combinations) × 6 replicates = N = 48 runs total, conducted across two experimental sessions (Day 1: n = 24; Day 2: n = 24).

Factor	Low Level (−1)	High Level (+1)
A: Detergent Type	Dishwashing (D)	Laundry (L)
B: Water Temperature	25°C (ambient)	60°C (heated)
C: Soaking Time	5 min	30 min

2.4 Randomization

Fabric pieces were physically shuffled before numbering (Section 2.3), so which piece occupied the n-1 vs. n-2 position within each twin set was randomly determined. Since Container 1 (ambient) was always assigned to n-1 and Container 2 (heated) to n-2 (Section 3), this shuffle constitutes the randomization mechanism for the temperature factor.

Soaking time and detergent type were assigned using Excel's =RAND() function: each twin set was ranked by its generated random value, and the resulting rank determined the treatment levels for these two factors. Twin sets were then processed in sequential numerical order (Set 1 through Set 12) within each day; because treatment assignment itself was independently randomized, processing order does not confound treatment with time-of-day effects.



2.5 Blocking

Day block (between-session): Because Day 2 used a different fabric batch and was conducted on a different day under potentially different ambient conditions, Day was included as a blocking factor in the statistical model. Day 1 and Day 2 each contributed 24 observations (3 replicates per treatment combination). The Day block effect was statistically significant ($F(1,39) = 4.41$, $p = 0.042$), confirming that blocking was warranted.

3. Nuisance Variable Control

- **Sauce fixation time:** Fixed at exactly 30 minutes for all pieces.
- **Mechanical agitation:** Standardized at 80 BPM using a metronome for 3 minutes; experimenter switched hands at 1 minute 30 seconds to eliminate directional bias.
- **Single experimenter:** All washing and agitation steps performed by one person to eliminate inter-experimenter variance.
- **Photography:** Fixed distance (100 cm), locked Pro-mode settings, same closed bathroom, same lighting, and same time of day for all trials.
- **Detergent concentration:** Both detergents prepared at an identical concentration of 1.44% w/w (5 g detergent per 347.4 g total solution).
- **Twin-set Pairing:** Pieces sharing the same index number (n-1 and n-2) form a twin set (Sets 1–12). Both pieces receive sauce simultaneously to control for variability in sauce application. The n-1 piece is assigned to Container 1 (ambient water) and the n-2 piece to Container 2 (heated water).

4. Experimental Procedure

1. Cut the T-shirt into 24 pieces of 5 cm × 5 cm; shuffle and number sequentially.
2. Photograph all pieces in the standardized setting (pre-contamination baseline).
3. Start stopwatch; apply 5 g of Buldak sauce simultaneously to each twin set (n-1 and n-2); record application time.
4. Allow sauce to fix for exactly 30 minutes.
5. Measure and record temperatures of Container 1 (ambient, 25°C) and Container 2 (heated, 60°C).
6. At 30 minutes post-application, submerge n-1 in Container 1 and n-2 in Container 2.

7. Per the random-number table: numbers 1–6 → 5-min soak; numbers 7–12 → 30-min soak.
8. Prepare detergent solution (5 g + 342.4 g water; type assigned per random-number table). Add to container; agitate for 3 minutes at 80 BPM, switching hands at 1:30.
9. Rinse under running water for 3 minutes.
10. Dry all pieces in a single room overnight.
11. Photograph dried pieces the following day under identical conditions (post-wash).

5. Sample Size and Power Analysis

Residual standard deviation: σ = 1.685 ΔE^*_{00} units, taken from the MSE of the full factorial ANOVA model with Day blocking (MSE = 2.840, df = 39).

Minimum detectable effect (MDE): According to the CIEDE2000 perceptual scale, a color difference of 2.0 ΔE^*_{00} units corresponds to the threshold at which two fabric samples become clearly distinguishable to the naked eye. We define this as the minimum practically meaningful effect: a laundering condition that fails to produce a 2.0-unit improvement relative to another condition would not be perceptible to a consumer.

Power computation: Computed in R using the non-central F distribution (pf(..., ncp = λ, lower.tail = FALSE)), with non-centrality parameter $\lambda = 2 \times n\_per\_level \times (MDE/2)^2 / \sigma^2$. The Day block consumes 1 degree of freedom, giving $df_2 = 39$.

MDE (ΔE^*_{00})	Perceptual meaning	λ	Power (N=48, α=0.05)
1.0	Barely perceptible	4.23	0.518
1.5	Intermediate	9.51	0.852
2.0 (target)	Clearly perceptible	16.90	0.980

Conclusion: Under the blocked model, with N = 48, the design achieves power = 0.980 at $\alpha = 0.05$ for detecting a practically meaningful effect of MDE = 2.0 ΔE^*_{00}, well above the conventional threshold of 0.80. This indicates that the design had sufficient sample size to detect effects of this magnitude, had they been present.

6. Ethics and Safety

The experiment was designed to be fully safe and to require no external ethical review.

- **Participants:** No external human participants; only fabric pieces are used as experimental units.
- **Contaminant and detergents:** Buldak sauce, dishwashing detergent, and laundry detergent are all commercially available household products with no chemical hazard at the quantities used.
- **Heated water:** Handled using thermally insulated gloves and a ladle to prevent direct skin contact.
- No consent procedure, IRB review, or contingency plan is required.

7. Execution

7.1 Deviations from the Progress Report Protocol

- **Response variable direction corrected:** Progress report feedback identified an inconsistency in the stated direction of ΔE^*_{00}. This has been resolved throughout: a smaller ΔE^*_{00} indicates better stain removal.
- **Day block added:** The additional 24 observations (Day 2) required a new fabric batch and were conducted on a different day. Day was incorporated as a blocking factor in the statistical model.
- **Power analysis inputs revised:** Progress report feedback noted that the effect size should balance what is realistic against what is practically meaningful. The MDE was redefined as 2.0 ΔE^*_{00} (clearly perceptible threshold on the CIEDE2000 scale) rather than the pilot-observed difference.
- **Model notation made consistent:** Main effect and interaction notation now uses a unified Greek-letter convention throughout (see Section 8.2).
- **Additive model rationale stated:** The additive model is fitted as a baseline to assess whether including interaction terms provides a meaningful improvement.

7.2 Protocol Adherence

Both experimental sessions followed the prescribed protocol. The Day 2 fabric batch difference was anticipated and addressed through blocking. No other deviations occurred.

8. Data Analysis

8.1 Descriptive Statistics

Response: ΔE^*_{00} (CIEDE2000), computed in R via farver. Smaller values indicate more effective stain removal. Observed values ranged from approximately 19.3 to 27.2, with an overall mean of 22.1 ΔE^*_{00} units. No condition achieved complete stain removal ($\Delta E^*_{00} \approx 0$).

Factor	Level	Marginal Mean ΔE^*_{00} (N=48)	Interpretation
Detergent (A)	Dishwashing (D)	22.49	Higher (less effective)
Detergent (A)	Laundry (L)	21.77	Lower (more effective)
Temperature (B)	25°C (ambient)	22.44	Higher (less effective)
Temperature (B)	60°C (heated)	21.81	Lower (more effective)
Soaking Time (C)	5 min	21.92	Lower (more effective)
Soaking Time (C)	30 min	22.33	Higher (less effective)
Block	Day 1	22.64	
Block	Day 2	21.62	

8.2 Statistical Model

Day was included as a blocking factor (main effect only). The three treatment factors were modeled with all main effects and interactions up to the three-way:

$$\Delta E*_{00} = \mu + \delta_{Day} + \alpha_A + \beta_B + \gamma_C + (\alpha\beta)_{AB} + (\alpha\gamma)_{AC} + (\beta\gamma)_{BC} + (\alpha\beta\gamma)_{ABC} + \varepsilon$$

where $\varepsilon \sim N(0, \sigma^2)$ i.i.d., δ_Day is the block effect, and all treatment factors are fixed effects at two levels.

Twin-set pairing (Section 3) controlled for sauce-application timing but is not represented as a separate term in the model, since each pair spans only two of the eight treatment cells rather than constituting a complete block; Day is the only blocking factor included above.

Role of the additive model: To determine whether the data justify the added complexity of interaction terms, an additive model (Day + A + B + C) was first fitted as a parsimony baseline. Because the B×C interaction term proved statistically significant in the full factorial model (Section 8.3), the additive model was judged insufficient, and the full factorial model was retained as the primary basis for inference.

8.3 ANOVA Results

Additive model (main effects only, Day block included):

Source	df	SS	MS	F	p-value
Day (Block)	1	12.520	12.520	4.2161	0.046 *
Detergent Type (A)	1	6.303	6.303	2.1225	0.152
Temperature (B)	1	4.818	4.818	1.6224	0.210
Soaking Time (C)	1	1.981	1.981	0.6671	0.419
Residuals	43	127.692	2.9696		

Full factorial model (all main effects and interactions, Day block included):

Source	df	SS	MS	F	p-value
Day (Block)	1	12.520	12.520	4.409	0.042 *
Detergent Type (A)	1	6.303	6.303	2.220	0.144
Temperature (B)	1	4.818	4.818	1.697	0.200
Soaking Time (C)	1	1.981	1.981	0.698	0.409
A×B	1	0.597	0.597	0.210	0.649
A×C	1	2.448	2.448	0.862	0.359
B×C	1	13.738	13.738	4.838	0.034 *
A×B×C	1	0.165	0.165	0.058	0.811
Residuals	39	110.745	2.840		

The Day block effect is significant ($F(1,39) = 4.41$, $p = 0.042$), confirming that blocking was justified. All three main effects and all interaction terms involving detergent type are non-significant ($p > 0.14$). The sole statistically significant treatment effect is the Temperature × Soaking Time interaction ($F(1,39) = 4.84$, $p = 0.034$).

8.4 Key Finding: Temperature × Soaking Time (B×C) Interaction

The significant B×C interaction ($F(1,39) = 4.84$, $p = 0.034$) replicates the pilot finding ($p = 0.017$) with the full dataset, **strengthening confidence in the result.** The interaction is shown in Figure 1 and summarized in the cell means below:

	25°C (Ambient)	**60°C (Heated)**	**Effect of temperature**
5-min soak	22.78	21.07	−1.70 → heated water more effective
30-min soak	22.11	22.55	+0.44 → heated water slightly less effective
Effect of soaking time	−0.67 (longer → slightly better)	+1.48 (longer → worse)	

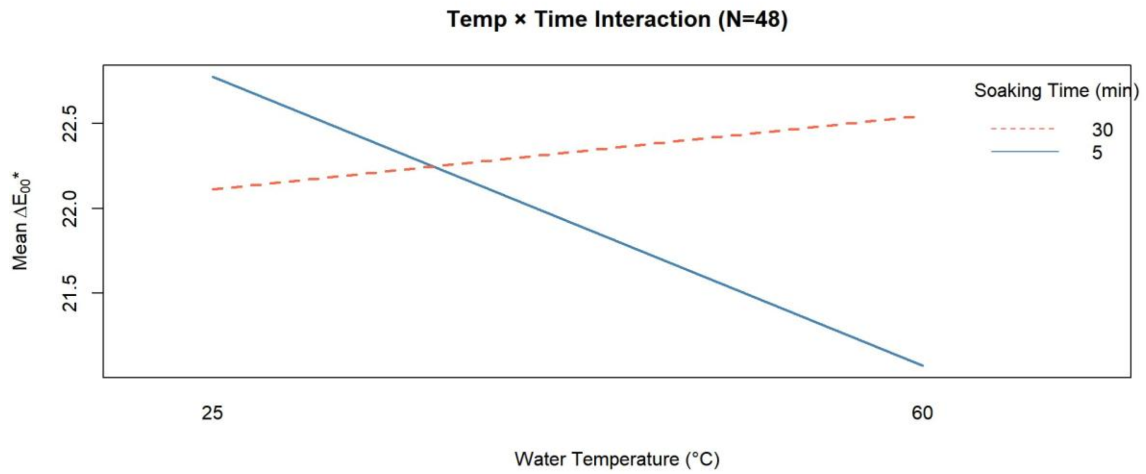


Figure 1. *Temperature × Soaking Time interaction plot (N=48). The crossing of the two lines indicates a significant disordinal interaction ($F(1,39) = 4.84$, $p = 0.034$): heated water (60°C) substantially improves stain removal at 5-minute soaking but has negligible or slightly negative effect at 30-minute soaking. Smaller ΔE^*_{00} = better removal.*

Although heated water (60°C) produced a numerically lower ΔE_{00} *than ambient water (25°C) at 5-minute soaking (21.07 vs. 22.78, a difference of 1.70 units) — potentially because elevated temperature accelerates emulsification of the chili-oil pigment before it can re-fix to the fabric — and a small reversal occurred at 30-minute soaking (22.55 vs. 22.11, a difference of 0.44 units), neither simple effect individually reaches the pre-specified practical significance threshold (MDE = 2.0).* The interaction contrast (2.14 ΔE_{00}) exceeds the 2.0-unit benchmark adopted in this study for practical significance, suggesting that the Temperature × Soaking Time interaction may also be practically meaningful.

The crossing pattern in Figure 1 is characteristic of a disordinal interaction and explains why neither the temperature main effect nor the soaking time main effect is statistically significant: their effects cancel at the marginal level because their directions differ depending on the other factor's level.

Treatment combination rankings (Figure 2):

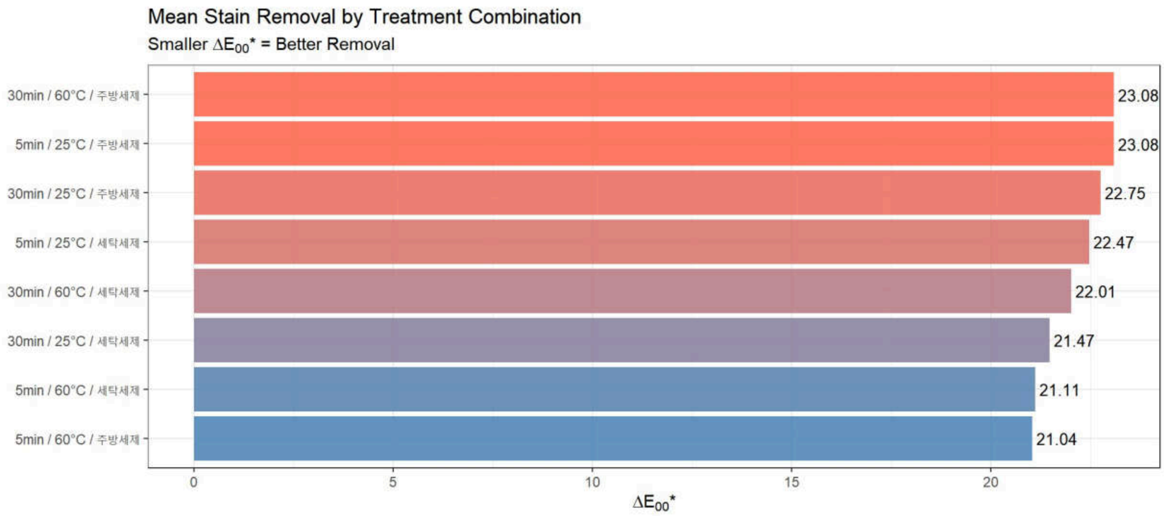


Figure 2. *Mean ΔE^*_{00} by treatment combination (N=48), ordered from most effective (smallest ΔE^*_{00}, blue) to least effective (largest, red). The two best conditions are both in the 5-min / 60°C group, regardless of detergent type.*

Figure 2 shows that the two best-performing conditions are 5-min / 60°C / dishwashing (ΔE_{00} = 21.04) and 5-min / 60°C / laundry (ΔE_{00} = 21.11), both belonging to the 5-min / 60°C group, consistent with the interaction pattern identified above. The worst conditions are 30-min / 60°C / dishwashing (23.08) and 5-min / 25°C / dishwashing (23.08). The maximum observed difference between best and worst conditions is approximately 2.1 ΔE^*_{00} units — at the boundary of the practically meaningful threshold (MDE = 2.0), suggesting that the choice of temperature–soaking time combination does matter in practice, though only marginally. However, because detergent type has no statistically significant effect ($p > 0.14$ for the main effect and all interactions involving Type), the ranking differences between detergents within the same Temperature × Soaking Time cell (e.g., 21.04 vs. 21.11) should be interpreted as sampling noise rather than a genuine treatment effect. The recommended optimal condition is therefore **5-minute soaking with 60°C water, regardless of detergent type**.

8.5 Day Block

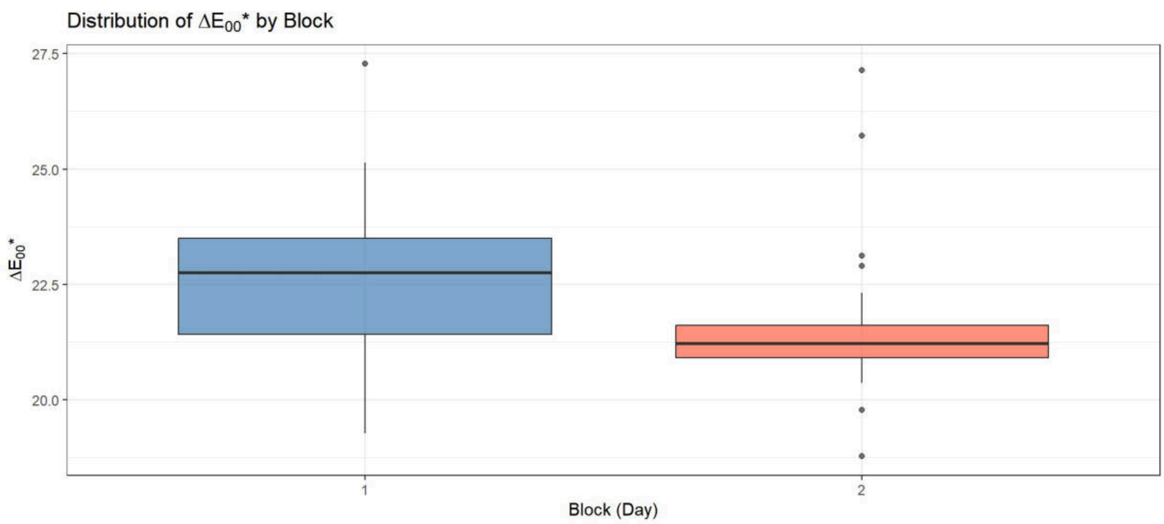


Figure 3. *Distribution of ΔE^*_{00} by block (Day). Day 1 (blue) shows a higher median and greater spread than Day 2 (red), consistent with the significant Day block effect ($F(1,39) = 4.41$, $p = 0.042$). The difference (approximately 1.0 ΔE^*_{00} unit) is attributable to the different fabric batch and ambient conditions on the two experimental days.*

Figure 3 confirms the Day block effect visually. Day 2 shows a tighter interquartile range than Day 1, though it also displays a greater number of flagged outliers (5 vs. 1) — a pattern that follows naturally from Day 2's smaller IQR, since the outlier threshold (1.5×IQR) scales with box width. The overall difference in central tendency (approximately 1.0 ΔE^*_{00} unit lower median for Day 2) is most likely attributable to the new fabric batch used in Day 2, which may have had slightly different fiber properties or whiteness.

8.6 Residual Diagnostics

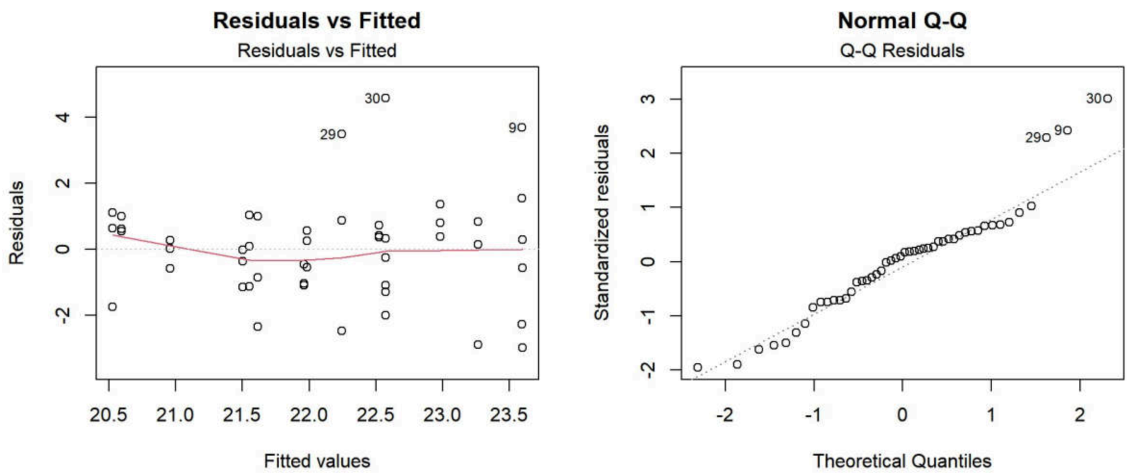


Figure 4. *Residual diagnostic plots for the full factorial model with Day blocking. Left: Residuals vs. Fitted values. Right: Normal Q-Q plot of standardized residuals.*

Figure 4 presents the residual diagnostic plots for the full factorial model with Day blocking. The Residuals vs. Fitted plot shows no strong systematic pattern, with the spread of residuals roughly constant across the range of fitted values — no evidence of a megaphone shape that would indicate nonconstant variance.

The Normal probability plot (NPP) shows that most residuals lie close to the reference line, with mild upper-tail curvature contributed by three observations (9, 29, 30). To formally assess whether these are outliers, we computed the externally Studentized residuals and compared them to the Bonferroni-corrected critical value $t_{0.05}/(2\times48), 39 = 3.551$. The largest externally Studentized residual (observation 30) is 3.391. No externally studentized residual exceeded the Bonferroni-adjusted critical value.

8.7 Temporal Dependence

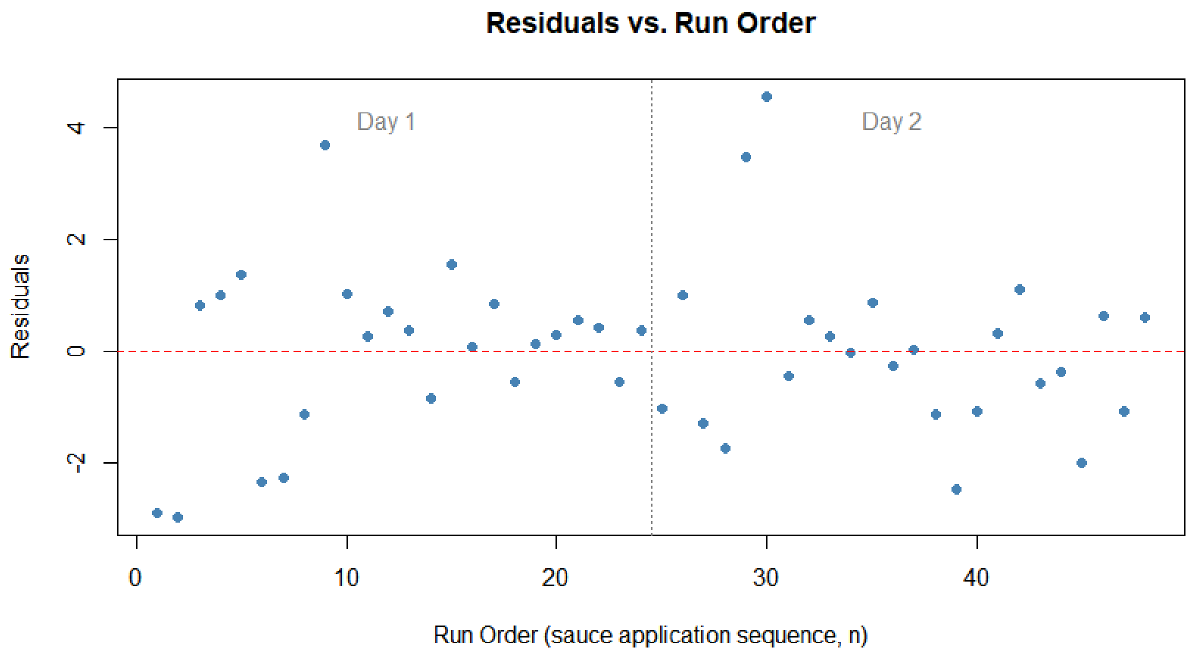


Figure 5. *Residuals plotted against run order (sequence of sauce application, n), separated by Day. No strong runs of same-signed residuals are evident, providing visual support for the independence assumption.*

Because pieces were processed sequentially in the order of sauce application, we checked for serial dependence by plotting residuals against run order (Figure 5) and computing the Durbin-Watson statistic. The index plot shows no strong runs of same-signed residuals across the run order. The Durbin-Watson statistic is DW = 1.532 ($p = 0.052$, one-sided test for positive autocorrelation), which falls just within the conventional acceptable range of 1.5–2.5, though close to its lower boundary. This provides weak, inconclusive evidence of mild positive serial correlation, but it does not reach statistical significance at $\alpha = 0.05$. Given the borderline nature of this result, we do not consider it strong enough to invalidate the independence assumption, though it is noted as a minor limitation (Section 9.2).

9. Conclusions

9.1 Main Findings

- **The Temperature × Soaking Time interaction is the only statistically significant treatment effect ($F(1,39) = 4.84$, $p = 0.034$),** replicated consistently across both the pilot (N=24, p=0.017) and the full dataset (N=48). This is the primary finding of the experiment.
- **Optimal laundering condition:** Heated water (60°C) combined with 5-minute soaking produces the best stain removal ($\Delta E^*_{00} \approx 21.0$–$21.1$), regardless of detergent type. Under time constraints, using heated water is strongly recommended.
- **Detergent type has no statistically significant effect** on stain removal under any of the tested conditions ($p = 0.144$). Either dishwashing or laundry detergent may be used.
- **All main effects are non-significant,** consistent with the disordinal nature of the B×C interaction: the directional reversal of effects causes cancellation at the marginal level.

- **Day blocking was statistically justified ($p = 0.042$),** confirming that fabric batch and ambient conditions between experimental sessions introduced meaningful nuisance variation.

9.2 Limitations and Future Directions

- The findings are specific to Buldak sauce on white cotton fabric. Different contaminants (e.g., kimchi, curry, coffee, or Malatang stains) and different fabric materials (e.g., polyester, linen, or blended fabrics) may respond differently to detergent type, water temperature, and soaking time. Therefore, the conclusions should not be generalized beyond the stain–fabric combination studied here.
- The Durbin–Watson statistic ($DW = 1.532$, $p = 0.052$) showed weak, borderline evidence of positive serial correlation in run order. Although this result does not reach statistical significance at $\alpha = 0.05$ and is not considered sufficient to invalidate the independence assumption, it remains a minor caveat that should be acknowledged.
- The experiment simulated hand-washing under controlled conditions. Results may not generalize directly to machine-washing, which involves substantially different levels of mechanical agitation, water circulation, and temperature control.
- The RGB-to-L*ab** conversion was based on calibrated smartphone photographs and assumed the standard sRGB color space. Although extensive efforts were made to standardize lighting and calibration, the exact color profile of the smartphone camera was not independently verified. Consequently, some systematic measurement error may remain.
- Each treatment factor was evaluated at only two levels. While the 2^3 factorial design efficiently identifies main effects and interactions, it cannot determine whether intermediate settings (e.g., 40°C water or 15-minute soaking) might yield superior stain-removal performance. Future studies could employ response-surface methods to identify the optimal operating conditions more precisely.
- Future work could also investigate additional factors not considered here, such as detergent concentration, enzyme-based pre-treatment products, or alternative stain-removal agents. Extending the experiment to other red-pigmented food stains would further improve the generalizability of the findings.